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Q1.(Basic Function)

```
def welcome():
    print("Welcome to Python Programming!")
welcome()
```

Q2.(Function with Parameter)

```
def greet(name):
    print("Hello,", name + "! Welcome back.")
greet("Vaishnavi")
greet("Vaishnavi")
```

Q3.(Default Parameter)

```
def show_message(message="Hello"):
    print(message)
show_message()
show_message("Welcome to Python!")
```

Q4.(Function with Multiple Parameters & Return)

```
def calculate_sum(a, b):
    return a + b

def print_sum(a, b):
```

```
    print("Sum =", a + b)
result = calculate_sum(10, 20)
print("Returned Sum =", result)
print_sum(10, 20)
```

Q5.(Positional vs Keyword Arguments)

```
def student_info(name, age):
    print("Name:", name)
    print("Age:", age)
student_info("Vaishnavi", 19)
student_info(age=20, name="shruti")
```

Q6.(Function with Return Value)

```
def square(num):
    return num * num
def cube(num):
    return num * num * num
number = int(input("Enter a number: "))
print("Square =", square(number))
print("Cube =", cube(number))
```

Q7.(Recursion - Basic)

```
def countdown(n):
    if n == 0:
        return
    print(n)
    countdown(n - 1)
countdown(10)
```

Q8.(Recursion - Factorial)

```
def factorial(n):
    if n == 0 or n == 1:
        return 1
    return n * factorial(n - 1)
num = int(input("Enter a number: "))
print("Factorial =", factorial(num))
```

Q9.(Scope - Local vs Global)

```
total = 0
def add_value(x):
    global total
    total += x
    local_var = x * 2
    print("Local Variable =", local_var)
```

```
add_value(10)
print("Total =", total)
add_value(20)
print("Total =", total)
add_value(30)
print("Total =", total)
```

Q10.(Mini Project - All Concepts)

```
def get_number():
    return int(input("Enter a number (0 to exit): "))
def is_even(num):
    return num % 2 == 0
def power(base, exp=2):
    return base ** exp
def show_result(num):
    if is_even(num):
  print("The number is Even.")
    else:
print("The number is Odd.")
    print("Square =", power(num))
while True:
    number = get_number()
    if number == 0:
        print("Program Ended.")
```

```
    break
show_result(number)
```